

Experiment 1 — Installation of Hypervisors and initiation of VMs with image file.

Install and configure an open-source hypervisor and create/instantiate a Virtual Machine (VM) using an existing image file. Demonstrate the successful installation and execution of the VM.

Tasks

- Install an open-source hypervisor such as Oracle VirtualBox
- Obtain a suitable VM image/ISO file.
- Create a new VM using the hypervisor.
- Configure the VM's CPU, RAM, storage, and network.
- Attach the image/ISO file to the VM.
- Start and boot the VM.
- Complete the guest OS installation, if required.
- Verify that the VM is running successfully.
- Installing the required build tools and Linux kernel headers using the below commands:

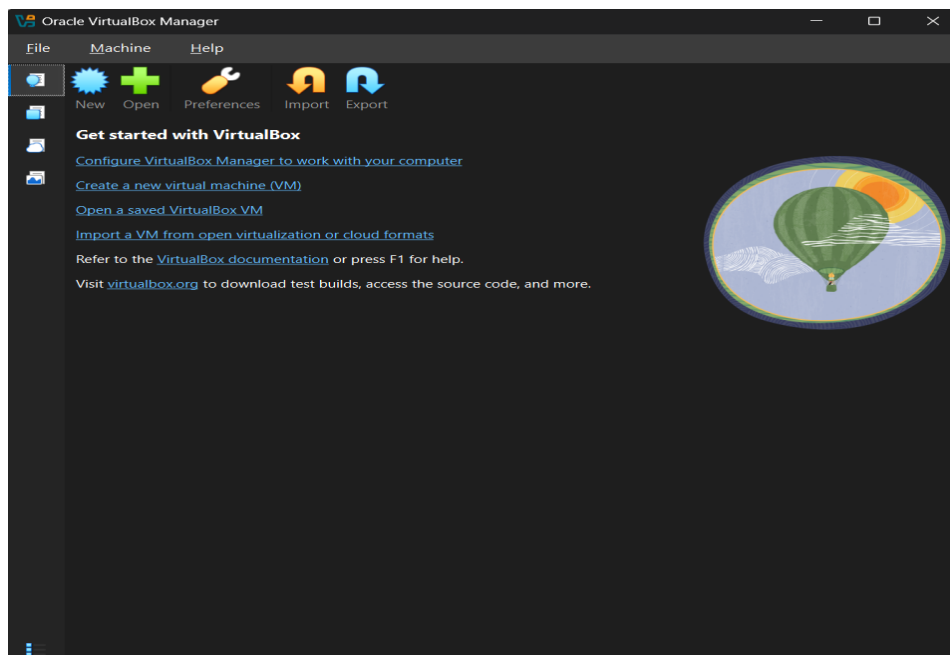
`sudo apt update`

`sudo apt install -y build-essential linux-headers-$(uname -r)`

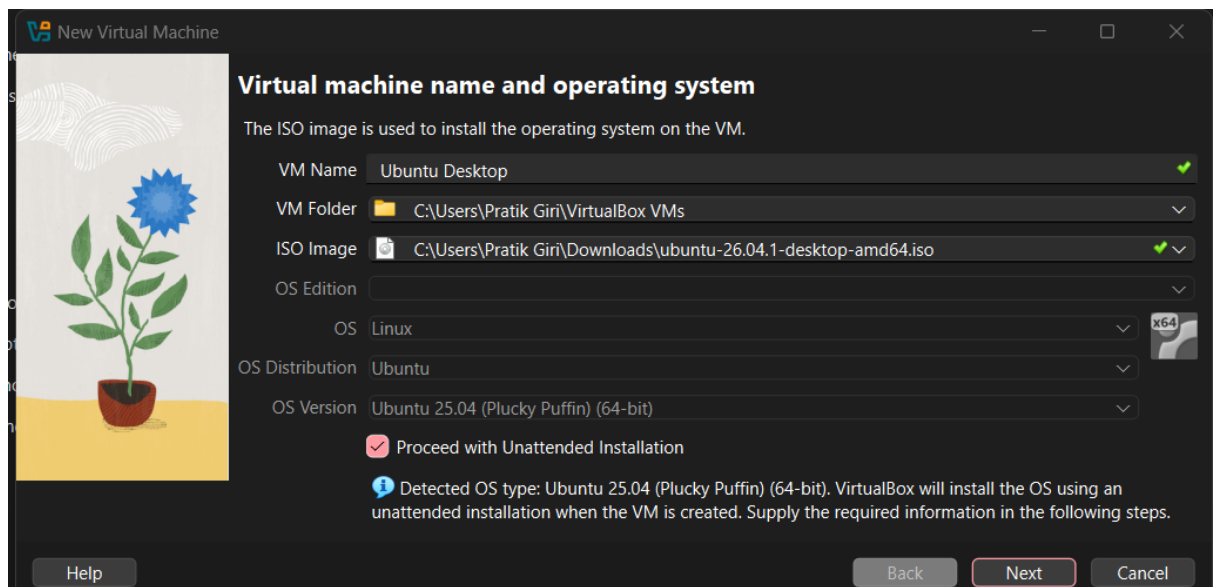
- Record the configuration and capture relevant screenshots.



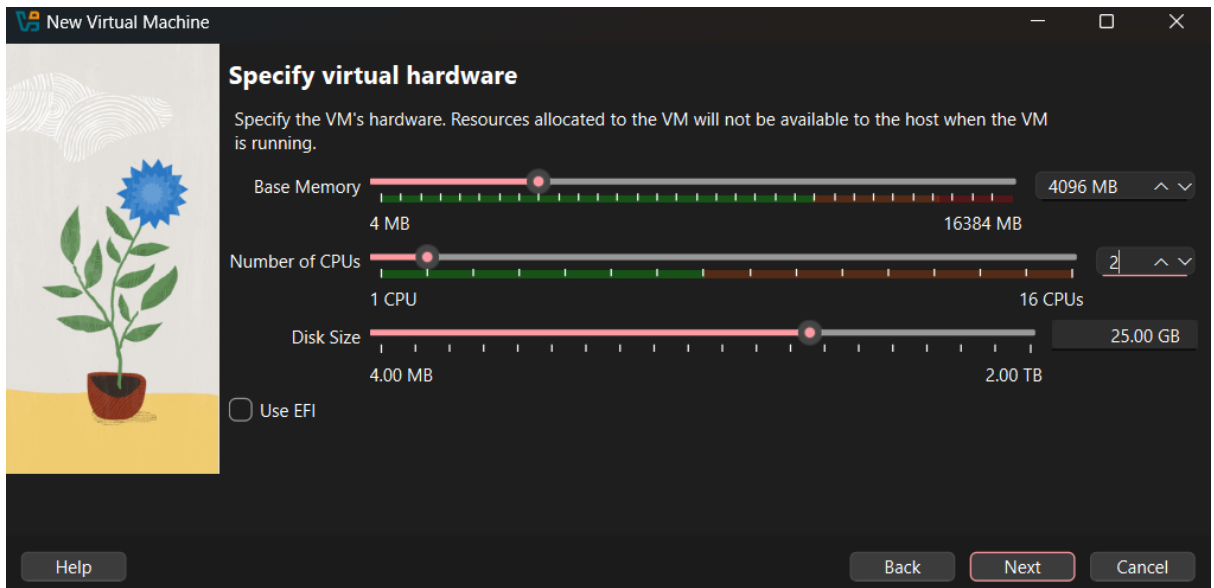
1. Installing and opening Oracle Virtual Machine



2. Uploading ISO file for OS installation in VM



3. Virtual Machine Configuration



Specify virtual hardware

Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.

Base Memory: 4096 MB (Range: 4 MB to 16384 MB)

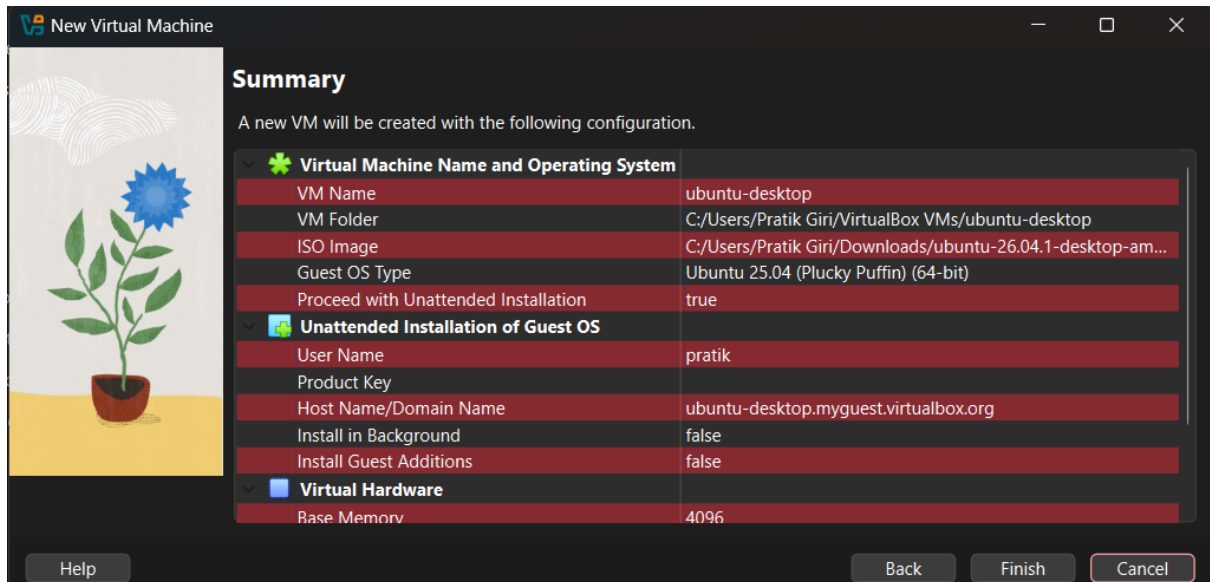
Number of CPUs: 2 (Range: 1 CPU to 16 CPUs)

Disk Size: 25.00 GB (Range: 4.00 MB to 2.00 TB)

☐ Use EFI

Buttons: Help, Back, Next, Cancel

4. Virtual Machine Configuration



Summary

A new VM will be created with the following configuration.

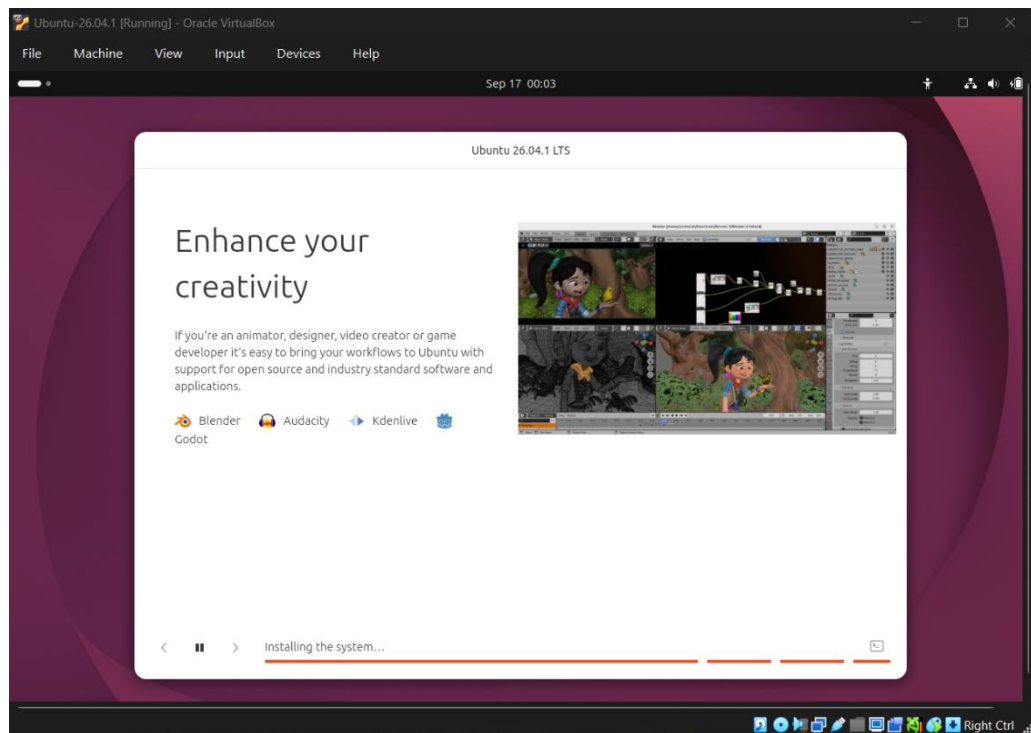
Virtual Machine Name and Operating System	
VM Name	ubuntu-desktop
VM Folder	C:/Users/Pratik Giri/VirtualBox VMs/ubuntu-desktop
ISO Image	C:/Users/Pratik Giri/Downloads/ubuntu-26.04.1-desktop-am...
Guest OS Type	Ubuntu 25.04 (Plucky Puffin) (64-bit)
Proceed with Unattended Installation	true

Unattended Installation of Guest OS	
User Name	pratik
Product Key	
Host Name/Domain Name	ubuntu-desktop.myguest.virtualbox.org
Install in Background	false
Install Guest Additions	false

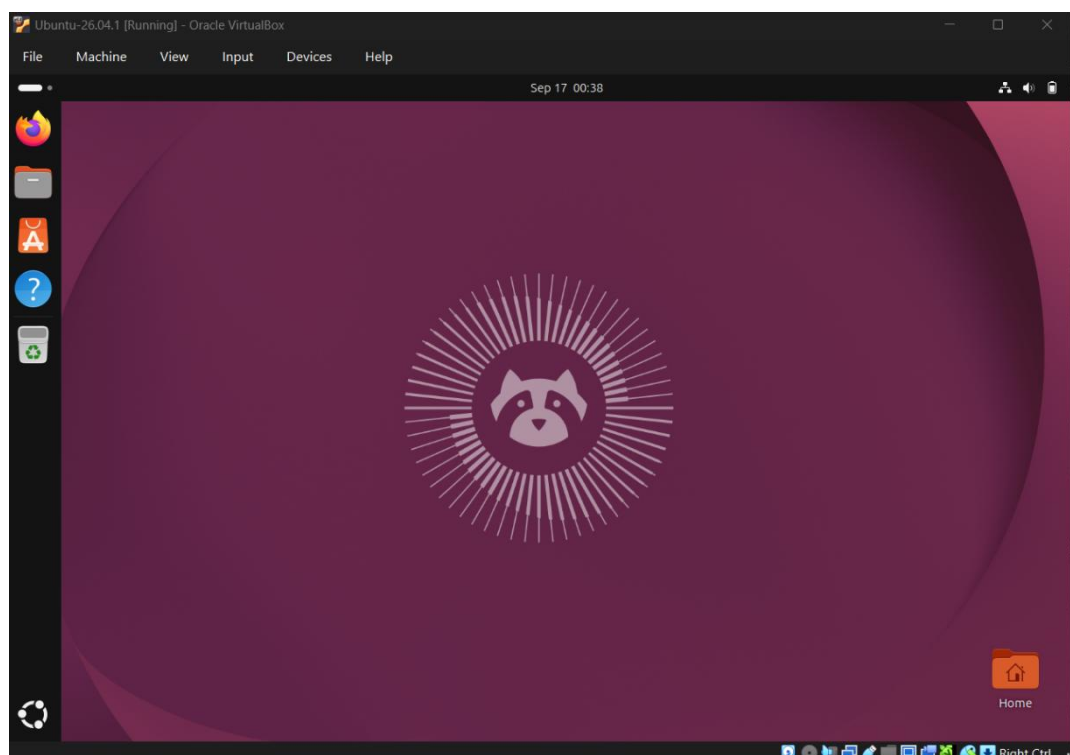
Virtual Hardware	
Base Memory	4096

Buttons: Help, Back, Finish, Cancel

5. Ubuntu installation in VM

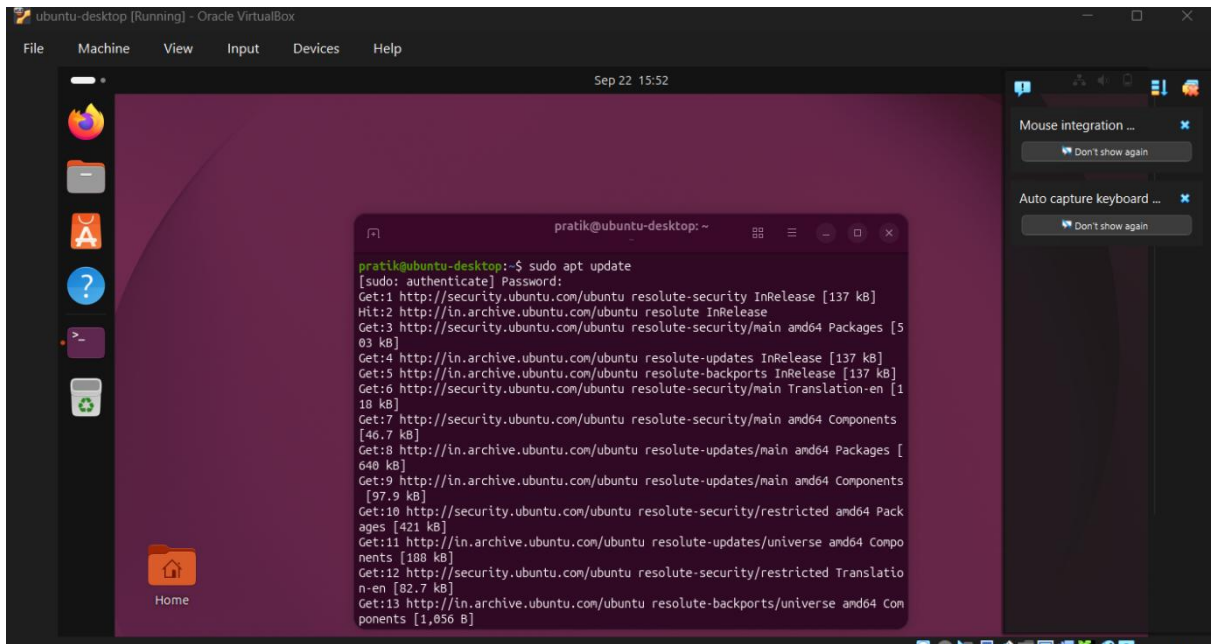


6. Ubuntu Home Screen



7. Install build tools & kernel headers using following command

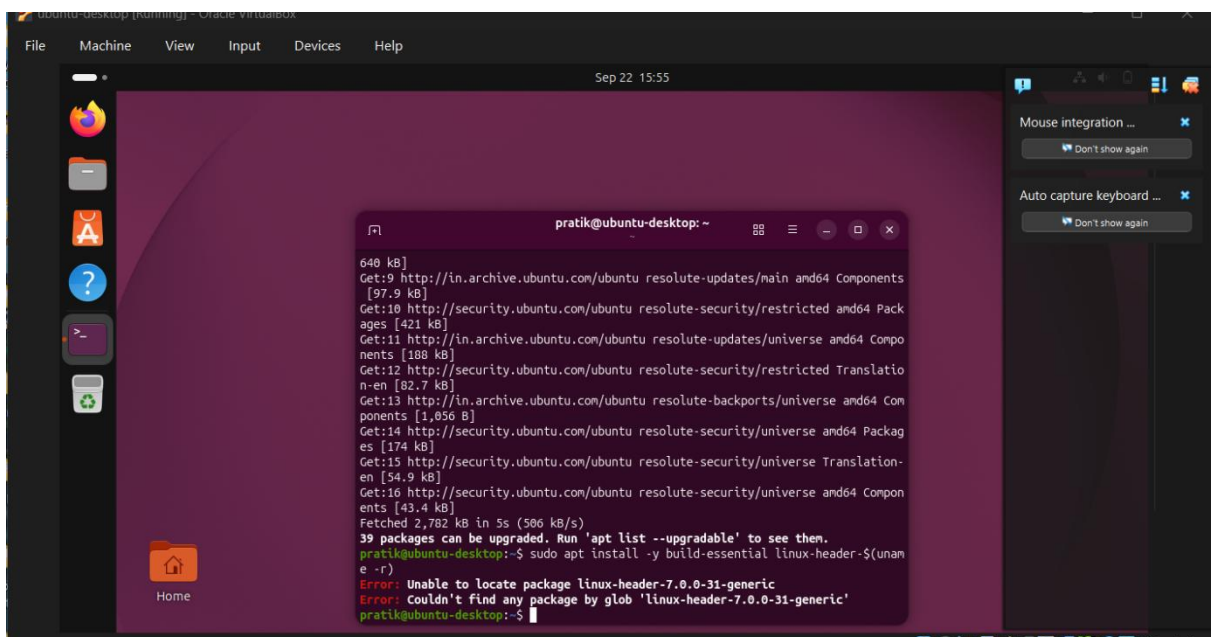
a) `sudo apt update`



The screenshot shows a terminal window titled 'pratik@ubuntu-desktop: ~' within an 'ubuntu-desktop [Running] - Oracle VirtualBox' environment. The terminal displays the output of the command `sudo apt update`. It starts with a password prompt and then lists various package sources and their sizes, including security updates, resolute updates, and restricted packages. The output shows that 39 packages can be upgraded.

```
pratik@ubuntu-desktop:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [503 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [118 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [640 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.9 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [421 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [82.7 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
```

b) `sudo apt install -y build-essential linux-headers-$(uname -r)`



The screenshot shows a terminal window titled 'pratik@ubuntu-desktop: ~' within an 'ubuntu-desktop [Running] - Oracle VirtualBox' environment. The terminal displays the output of the command `sudo apt install -y build-essential linux-headers-$(uname -r)`. It shows the download progress of various packages, including build-essential and linux-headers. The output indicates that 39 packages can be upgraded and that the installation of build-essential and linux-headers-7.0.0-31-generic is successful.

```
pratik@ubuntu-desktop:~$ sudo apt install -y build-essential linux-headers-$(uname -r)
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [174 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [54.9 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Fetched 2,702 kB in 5s (506 kB/s)
39 packages can be upgraded. Run 'apt list --upgradable' to see them.
pratik@ubuntu-desktop:~$ sudo apt install -y build-essential linux-headers-$(uname -r)
Error: Unable to locate package linux-header-7.0.0-31-generic
Error: Couldn't find any package by glob 'linux-header-7.0.0-31-generic'
pratik@ubuntu-desktop:~$
```